Experiment 5 Design of High Pass Filter using Windowing Techniques

5.1 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

Program Code:

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)
%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect');
%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');
%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');
%% --- Frequency Response Plots ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
```

title('High-pass FIR using Hamming Window');

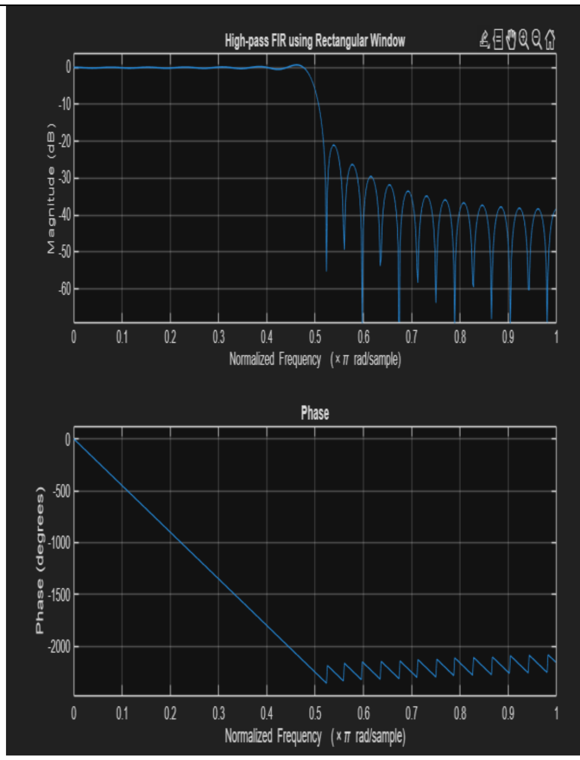
figure;

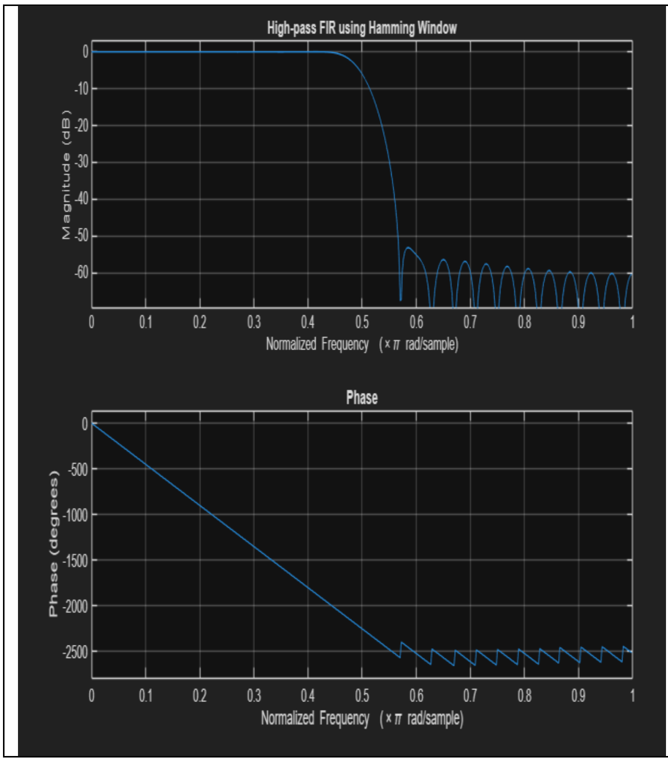
freqz(b_hann,1);

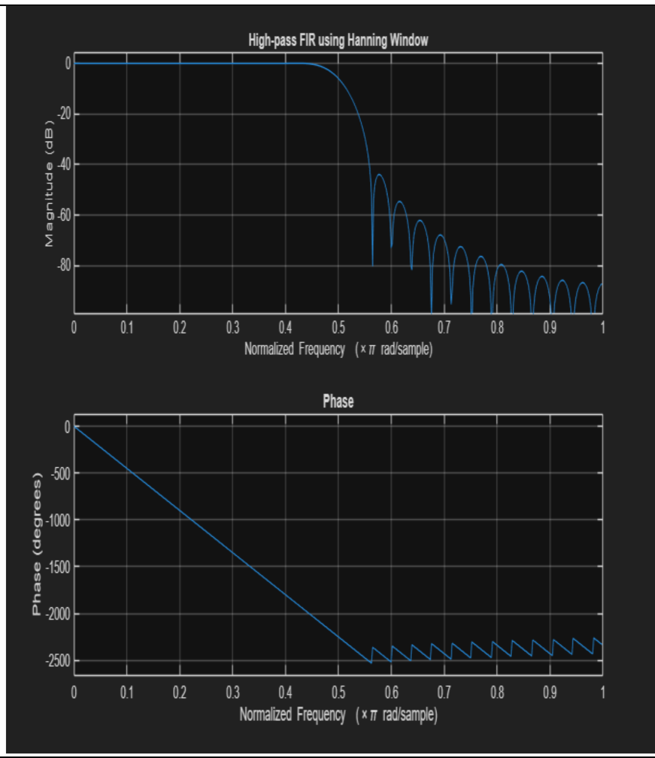
title('High-pass FIR using Hanning Window');

Output:

```
--- Rectangular Window Coefficients ---
    0.0126
   -0.0000
   -0.0137
    0.0000
    0.0150
   -0.0000
   -0.0166
    0.0000
    0.0185
   -0.0000
   -0.0210
    0.0000
    0.0242
   -0.0000
   -0.0286
    0.0000
    0.0349
   -0.0000
   -0.0449
    0.0000
    0.0629
   -0.0000
   -0.1048
    0.0000
    0.3145
    0.4940
    0.3145
    0.0000
   -0.1048
   -0.0000
    0.0629
    0.0000
   -0.0449
   -0.0000
```







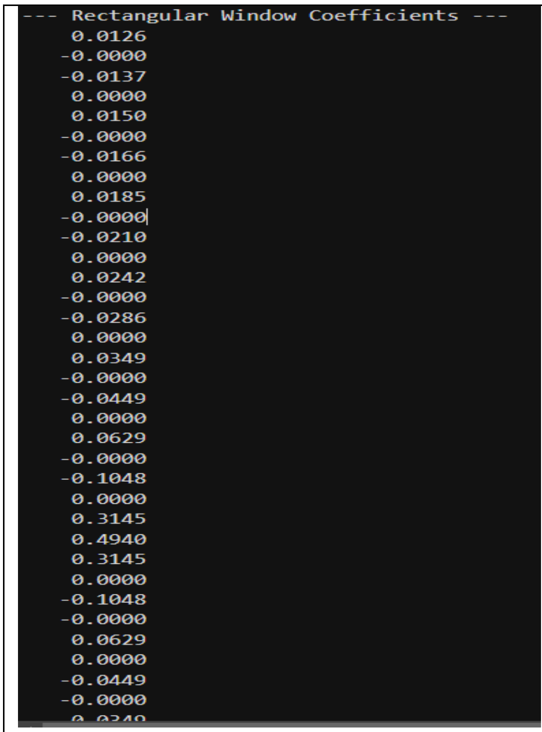
Experiment 5 Design of High Pass Filter using Windowing Techniques

5.2 Generate the coefficients of an FIR High-Pass filter using Rectangular windowing techniques.

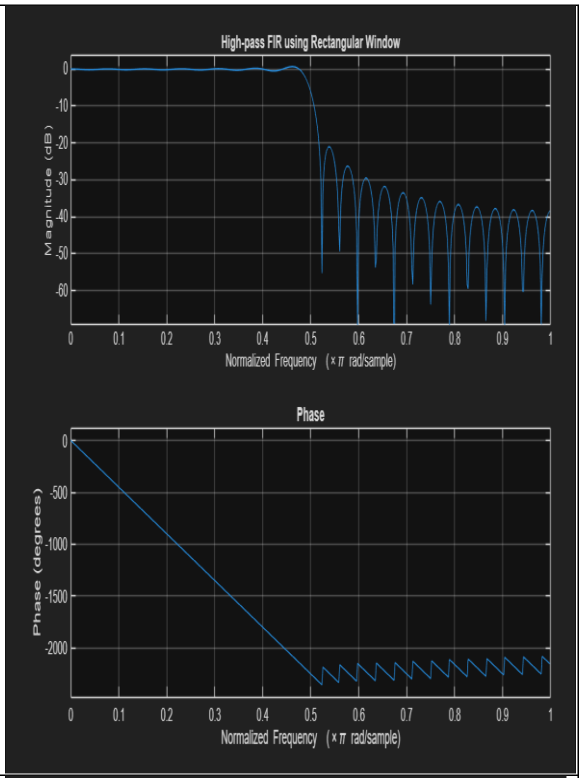
Program Code:

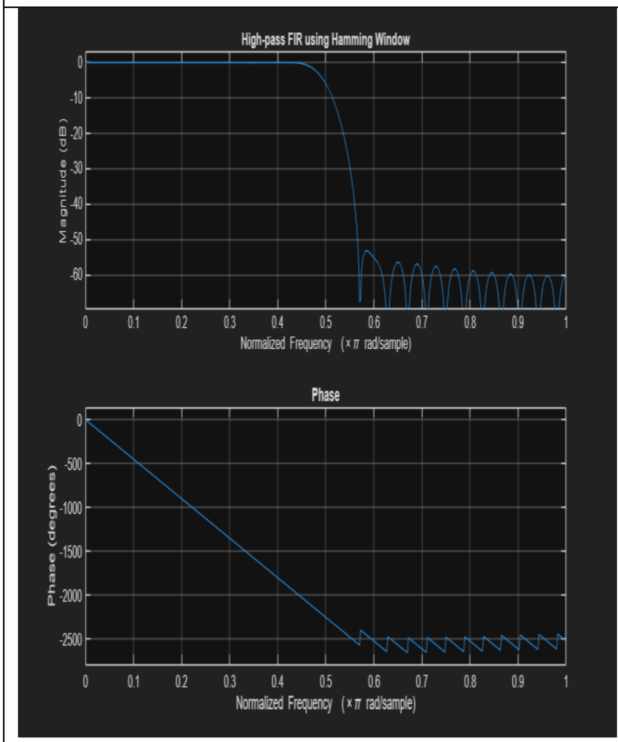
```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)
%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');
%% --- Frequency Response Plots ---
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
```

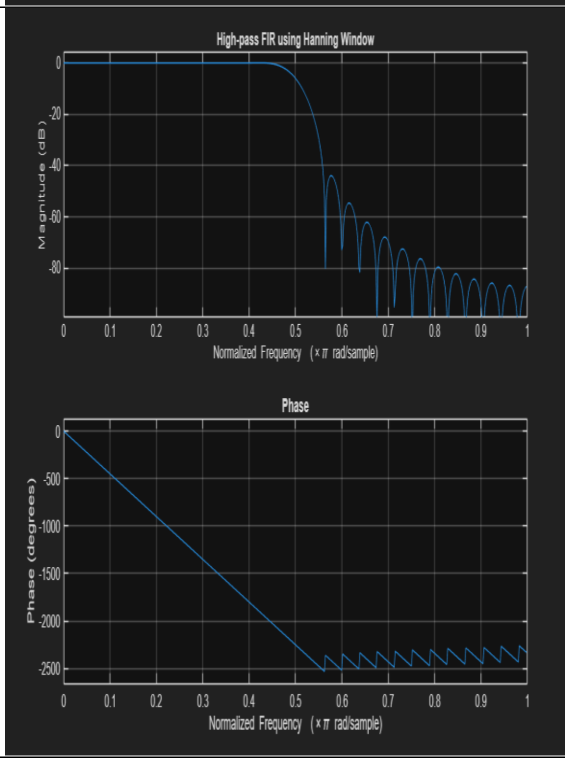
Output:



```
--- Rectangular Window Coefficients ---
    0.0126
   -0.0000
   -0.0137
    0.0000
    0.0150
   -0.0000
   -0.0166
    0.0000
    0.0185
   -0.0000
   -0.0210
    0.0000
    0.0242
   -0.0000
   -0.0286
    0.0000
    0.0349
   -0.0000
   -0.0449
    0.0000
    0.0629
   -0.0000
   -0.1048
    0.0000
    0.3145
    0.4940
    0.3145
    0.0000
   -0.1048
   -0.0000
    0.0629
    0.0000
   -0.0449
   -0.0000
```







Experiment 5 Design of High Pass Filter using Windowing Techniques

5.3. Generate the coefficients of an FIR High-Pass filter using Hanning windowing

Techniques

Program Code:

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0–1)
%% --- FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');
%% --- Frequency Response Plots ---
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```

Output:

```
--- High-Pass Coefficients (Hanning Window) ---
   -0.0000
   -0.0000
    0.0003
    0.0008
    0.0009
   -0.0000
   -0.0020
   -0.0038
   -0.0035
   -0.0000
    0.0057
    0.0100
    0.0087
   -0.0000
   -0.0127
   -0.0216
   -0.0183
   -0.0000
    0.0267
    0.0464
    0.0410
   -0.0000
   -0.0726
   -0.1568
   -0.2243
    0.7500
   -0.2243
   -0.1568
   -0.0726
   -0.0000
    0.0410
    0.0464
    0.0267
   -0.0000
```